Drive:

Dynamixel XL-330 Motor, roller chain

- Motor Constraints:
    - aiming for ~ 0.5 N.m usable torque out of the motor, but friction and detachment force are still unknown
- Roller Chain:
    - Length: 12 suction cups * suction cup width (2.77 cm) + 12* spacing (0.5-1cm)
    - Width: Fit 4 cups at bottom of belt
    - Height: aim for ~> height of valve
    - Keep chain greased/lubricated for performance
    - Screw abt 35g

1 or 2 motors along the bottom of the roller chain?

- Depends on detachment force + friction of roller chain, we previously assumed 1 would be enough
- Before we thought to have some slack in the chain to be able to handle curved surfaces, in this case 2 drive sprockets could be good (can be done with 1 motor as well)

Symmetric drive system (forwards and backward locomotion)(4 points of tension excluding belt tensioner)? Or single direction(3 points of tension)?

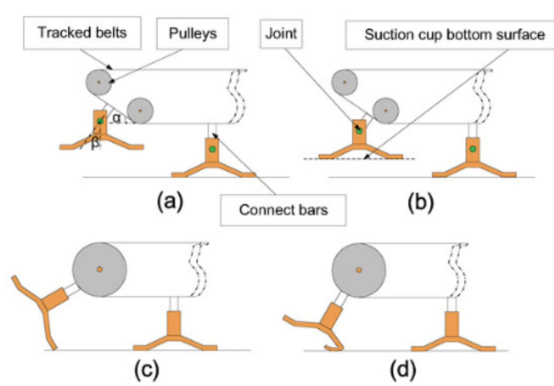
- For symmetric, unifying detachment/attachment methods in both directions might be difficult

How to support sprockets so that they can withstand the bending torque while also not interfering with tubing/vaccuum system?

- Sandwich gears individually or an entire sideplate with center hole for tubing

How do we deal with the angle of the suction cup attachment?

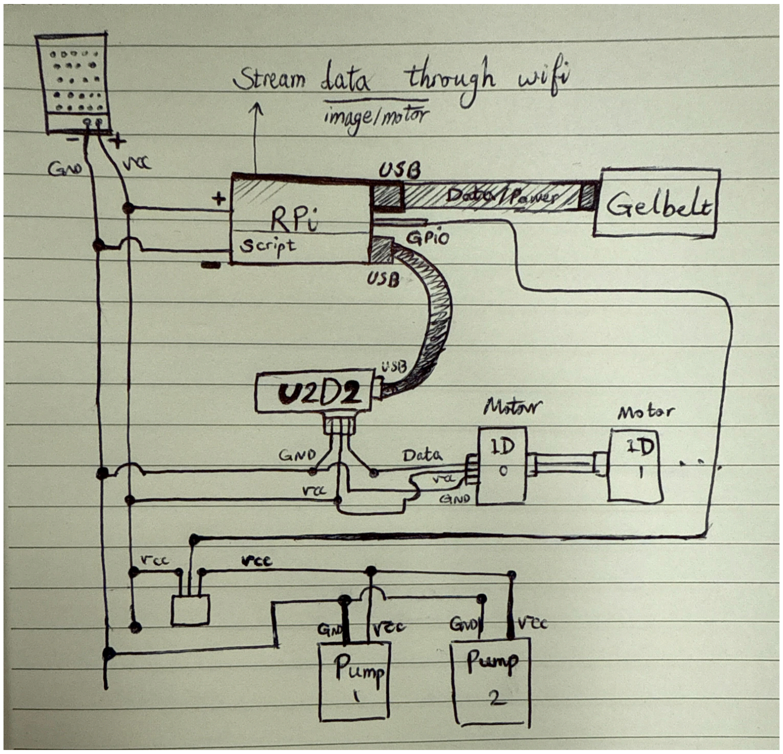
- Fixed angle
- Large turning radius top pulley / shallow approach angle
- Joint + polarity of magnets



Belt Torsion

- Deep socket guide rail
    - Limits our ability to go over curved surfaces as initially planned

Electronics:



03/20/2026 Electronics Initialization

IMG_3369.MOV

IMG_3384.MOV Revised for two separate codes for pump & motor + gelbelt

- leahlee2@172.16.125.4 (pwd leahlee1234)

  climbingbot.py

  Dynamixel Motor Control:
  m[number]   example: m100 or m0
  ml[number]   left motor only
  mr[number]   right motor only

  Camera Recording Control:
  g1 / g0   camera start / stop mp4 save

  q   quit

  pump.py
  p1 / p0   pump on / off

  q   quit

Dynamixel XL-330, Raspi, U2D2 (No powerhub board), 2 Vacuum Pump, 5V smps / offboard power
Estop motor switch?

- Do we need to get another pump?

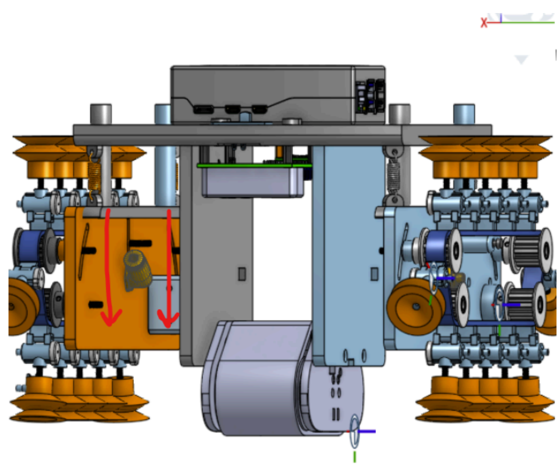
  stream data through wifi?

What is placement of all electronics on system? Weight Distribution?

- On top or under top plate?
  - Raspi width > width of gelbelt
  - U2D2 and vacuum pump is small enough to be on side or under

Suspension:

- K = 9.19 N / cm with a 1.5 cm offset between bottom of gelbelt and bottom of suction cup



Unbulk/reduce system size

- Move linear slides that press the Gelbelt into the surface to bottom of sideplate

Integrate sideplate with gelbelt sideplate

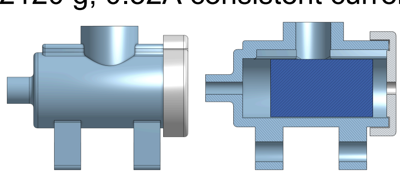
- how does this work with suspension?

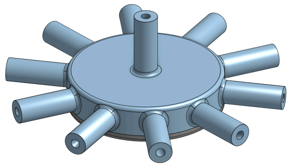
Detachment:

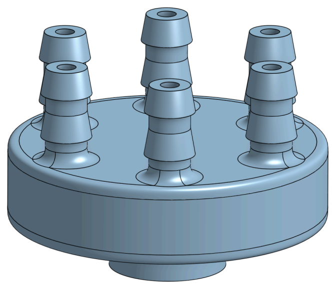
Pneumatic Valve:

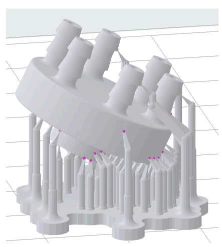
- Valve + peeling w/ string?
- Strong aluminum/steel shaft

3/14/2026 1 Minute 2 kg Pneumatic System Testing
https://drive.google.com/file/d/1Wm4aUn_hWcE9u1Hol4z5saehdakkWtDu/view?usp=sharing

- tested the pneumatic system for a minute to validate if there would be a noticeable air leak during continuous operation
  - 2120 g, 0.52A consistent current
  - 
    - Pneumatic valve used
    - Changes needed: chain connection part & wider suction cup connection part for vented screws

- 
    - Junction used (PLA printed)
    - Changes needed: 90 degree angled connection part for radial tube orientation in driving system (picture to be added).



- Modified Junction:
    - Printed with clear V4, no internal support, with 20 degrees angle
        - 
    - https://drive.google.com/file/d/1A1NTmomgjNOGU-j_HvkufhnD_d_WBWBw/view?usp=sharing
    - Modified Junction testing: 0.47A, 2120g.

Magnets:

- All around the track, one polarity to push the magnetic ball into the silicon seal and one polarity to push the magnet out at the bottom of the track
- I just tested the vacuum valve with new design. A magnet with 4.03lbs pull in the valve and a magnet with 4.65lbs pull outside is enough for making the vacuum and pulling the magnet out from vacuum. This is good since we are ordering a magnet with 4.7lbs pull
- https://www.kjmagnetics.com/magnet-strength-calculator.asp?srsltid=AfmBOooTzIO_YL64fIUrKw2z3WHo8rBhNJ2_zCEOjorYILOJBzI150gJ

- 1 cm magnet testing (1 suction cup)
    - 1 magnet 600 g
    - 2 magnets 1.2 kg
    - 3 magnets 1.5 kg
- 3 magnets 2 cm 700 g